

Addis Ababa University
Department of Mathematics
Calculus II: Math 2022
Worksheet IV

1. Using the definition verify

a) $\lim_{n \rightarrow \infty} \frac{2n-1}{n+1} = 2$ b) $\lim_{n \rightarrow \infty} \frac{n^2}{2n^2+1} = \frac{1}{2}$ c) $\lim_{n \rightarrow \infty} (\sqrt{n-1} - \sqrt{n}) = 0$

2. Find $\lim_{n \rightarrow \infty} a_n$ if it exists

a) $a_n = \left(1 + \frac{3}{n}\right)^{n+1}$ b) $a_n = \sqrt[3]{n+2} - \sqrt[3]{n+1}$ c) $a_n = \left(\frac{n^2-1}{n}\right)^n$
d) $a_n = \frac{n}{n+1} \sin \frac{n\pi}{2}$ e) $a_n = \tanh n$ f) $a_n = \frac{1+2+3+\dots+n}{n^2+1}$

3. Determine whether the given sequence is monotonic or not, and convergent or not

a) $\left\{\frac{3^{n+1}}{5+3^n}\right\}_1^\infty$ b) $\left\{\frac{n!}{n^n}\right\}_0^\infty$ c) $\{\cos n\pi\}$ d) $\left\{\ln\left(\frac{3n}{n+1}\right)\right\}$ e) $\{\sqrt[n]{3^n+4^n}\}$

4. Determine the convergence or divergence of the sequence $\{a_n\}_{n=1}^\infty$ if

a) $a_n = \frac{\ln n}{n+1}$ b) $a_n = \left(1 + \frac{1}{n^2}\right)^{3n}$ c) $a_n = \sum_{k=0}^n 2^{-k}$

5. Find a) $\lim_{n \rightarrow \infty} \frac{1+3^n}{n+2^n}$ b) $\lim_{n \rightarrow \infty} \frac{1+(-1)^n}{e^{2n}}$ c) $\lim_{n \rightarrow \infty} \sqrt[n]{4^n+5^n}$

6. Let $\{a_n\}$ be a sequence with $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2+a_n}$, $n \geq 1$.

a) Show that $\{a_n\}$ is convergent b) Find $\lim_{n \rightarrow \infty} a_n$

7. Determine the convergence or divergence of each of the following series. If it converges, find the sum.

a) $\sum_{n=1}^\infty 2\left(\frac{3}{5}\right)^{n-1}$ b) $\sum_{n=0}^\infty \left(-\frac{\pi}{e}\right)^n$ c) $\sum_{n=1}^\infty \frac{3^{2n}}{2^{3n-1}}$
d) $\sum_{n=1}^\infty \frac{3}{n(n+2)}$ e) $\sum_{n=1}^\infty \frac{n}{(n+1)!}$ f) $\sum_{n=2}^\infty \left(\frac{1}{n(n-1)} - \frac{1}{\sqrt{n}}\right)$

8. Show that $\sum_{n=1}^\infty \ln\left(1 + \frac{1}{n}\right)$ diverges and $\sum_{n=1}^\infty \ln\left(1 - \frac{1}{n^2}\right)$ converges.

9. Test the convergence or divergence of

a) $\sum_{k=1}^\infty \frac{\arctan k}{k^2+1}$ b) $\sum_{n=3}^\infty \frac{\sqrt{n}}{n-2}$ c) $\sum_{n=1}^\infty \frac{n+1}{\sqrt{n^5+4}}$
d) $\sum_{n=1}^\infty \frac{5^n-2^n}{3^n+4^n}$ e) $\sum_{n=1}^\infty \frac{n^n}{3^n n!}$ f) $\sum_{n=1}^\infty \left(\frac{n^2+1}{2n^2-1}\right)^n$

10. Find all values of p for which the series $\sum_{n=2}^\infty \frac{1}{n(\ln n)^p}$ converges.

11. Test for absolute convergence, conditional convergence or divergence for each of the following.

a) $\sum_{k=2}^\infty \frac{(-1)^k}{\ln k}$ b) $\sum_{n=1}^\infty \frac{(-1)^n n}{5n+1}$ c) $\sum_{n=1}^\infty \frac{(-1)^{n-1} (n+10)}{n(n^2+1)}$
d) $\frac{1}{3} - \frac{1}{2} + \frac{5}{9} - \dots + \frac{(-1)^{n+1} (2n-1)}{3n} + \dots$

12. Determine the convergence or divergence of

$$\begin{array}{lll} \text{a)} & \sum_{n=1}^{\infty} \frac{(n+2)!}{n! 10^n} & \text{b)} \quad \sum_{n=1}^{\infty} \frac{(-1)^{n+1} n!}{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)} & \text{c)} \quad \sum_{k=1}^{\infty} \frac{3 \cdot 5 \cdot 7 \cdot \dots \cdot (2k+1)}{2 \cdot 5 \cdot 8 \cdot \dots \cdot (3k-1)} \\ \text{d)} & \sum_{k=1}^{\infty} \left(\frac{k!}{k^k} \right)^k & \text{e)} \quad \sum_{n=1}^{\infty} n^t e^{-n}, \quad t \in \mathbb{R} & \text{f)} \quad \sum_{n=1}^{\infty} \sum_{k=1}^n 2^{-k} \end{array}$$

13. Find the interval of convergence and radius of convergence of

$$\begin{array}{lll} \text{a)} & \sum_{n=0}^{\infty} \frac{(x-1)^n}{n+2} & \text{b)} \quad \sum_{n=1}^{\infty} \frac{2^n (-x)^n}{\sqrt{n+1}} & \text{c)} \quad \sum_{n=0}^{\infty} \frac{n}{4^n} (2x+1)^n \\ \text{d)} & \sum_{n=0}^{\infty} \frac{(-1)^n}{9^n} x^{2n} & \text{e)} \quad \sum_{n=2}^{\infty} \frac{x^n}{(\ln n)^n} & \text{f)} \quad \sum_{n=0}^{\infty} \frac{3^n}{n^2+1} (1-x)^{n+1} \end{array}$$

14. Show that $\ln\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{1}{3}x^3 + \frac{1}{5}x^5 + \dots\right)$, $|x| < 1$.

15. Find the Taylor series of f at the given value of c , and determine the radius of convergence of the resulting series

$$\begin{array}{ll} \text{a)} & f(x) = \cos x, \quad c = \pi \\ \text{b)} & f(x) = \ln x, \quad c = 2 \\ \text{c)} & f(x) = \sqrt{x}, \quad c = 1 \\ \text{d)} & f(x) = \frac{x}{1-x}, \quad c = 0 \end{array}$$

16. Find the Maclaurin's series expansion of

$$\begin{array}{lll} \text{a)} & f(x) = \frac{1}{2+x} & \text{b)} \quad f(x) = x^2 e^{-3x} & \text{c)} \quad f(x) = \sin^2 x \\ \text{d)} & f(x) = \sinh 2x & \text{e)} \quad f(x) = \frac{1}{(1-x)^3} & \text{f)} \quad f(x) = \tan^{-1}(x^2) \end{array}$$